Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 3 Numerical characteristics of random variables

Definition of variance

The variance measures the spread of the distribution around the mean. An r.v. with a zero variance is constant. The larger the variance, the more uncertain the r.v. is, in the mean square sense.

Definition of the variance If X is an r.v. with $E[X] = \mu_X$, then $Var(X)$ or $D(X)$, called the variance of X, is defined by

$$Var(X) = D(X) = E\left[(X - \mu_X)^2\right]$$

and the Standard Deviation (SD) σ_X of X is

$$\sigma_X = \sqrt{Var(X)} = \sqrt{D(X)}$$

The square is used to ensure that all differences X-E(X) have a positive effect

Also, one may use σ^2 or $\sigma_X{}^2$ for the variance of the r.v. X. The variance of X is defined as the average square distance between X and its mean μX . The standard deviation is used more typically since the units associated with SD are the same as the r.v. X itself, whereas the variance is measured in squared units of X .

Variance of discrete random variables

1. **Definition.** Suppose the distribution law of the discrete random variable X is:

X	x_0	x_1	$x_2 \cdots\cdots$	$x_n \cdots\cdots$
P_K	p_0	p_1	$p_2 \cdots\cdots$	$p_n \cdots\cdots$

If the series $\sum_{k=1}^{\infty}[x_k - E(X)]^2 p_k$ absolutely converges, then this series is called the variance of X, which is recorded as:

$$D(X) = Var(X) = \sum_{k=1}^{\infty}[x_k - E(X)]^2 p_k = E[X - E(X)]^2$$

Note. If the series $\sum_{k=1}^{\infty}[x_k - E(X)]^2 p_k$ does not absolutely converge, it is said that Var(X) does not exist.

Variance of continuous random variables

1. Definition.

Suppose the probability density of continuous random variable X is f(x).

If the integral $\int_{-\infty}^{+\infty} [x - E(X)]^2 f(x)dx$ is absolutely convergent, then this integral is called the variance of X, which is recorded as:

$$D(X) = Var(X) = \int_{-\infty}^{+\infty} [x - E(X)]^2 f(x)dx$$

$$= E[X - E(X)]^2$$

The prosperities of variance

1. $Var(X) = E(X^2) - [E(X)]^2$

This is an VERY important property, and frequently used calculation formula

Prove: $\because \quad Var(X) = E[X - E(X)]^2$

$$= E[X^2 - 2XE(X) + (E(X))^2]$$

$$= E(X^2) - E[2XE(X)] + E[(E(X))^2]$$

$$= E(X^2) - 2E(X)E(X) + [E(X)]^2$$

$$= E(X^2) - [E(X)]^2$$

Note: This formula gives a formula for calculating the variance of a random variable X, and also gives the relationship between mathematical expectation and variance.

Properties of the variance

2. If c is a constant, then: $Var(c)=0$

The sufficient and necessary condition for $Var(X)=0$ is that X takes the constant c with probability 1, that is, **P(X=*c*)=1**,

Obviously, here $c = E(X)$.

3. If c is a constant and X is a random variable, then:

$$Var(cX) = c^2 Var(X)$$

Prove: $Var(cX) = E[(cX)^2] - [E(cX)]^2$

$$= E(c^2X^2) - [cE(X)]^2$$

$$= c^2E(X^2) - c^2[E(X)]^2$$

$$= c^2[E(X^2) - (E(X))^2] = c^2Var(X)$$

Properties of the variance

4. If X, Y are independent random variables, then:

$$Var(X+Y) = Var(X) + Var(Y)$$

Because X and Y are independent of each other, X-E(X) and Y-E(Y) are also independent of each other

Prove: $Var(X+Y) = E[(X+Y) - E(X+Y)]^2$

$$= E\{[X - E(X)] + [Y - E(Y)]\}^2$$

$$= E[X - E(X)]^2 + E[Y - E(Y)]^2 + \underline{2E\{[X - E(X)]}\underline{[(Y - E(Y)]\}}$$

$$= Var(X) + Var(Y) \qquad +2E[X - E(X)] \cdot E[Y - E(Y)]$$

$$= Var(X) + Var(Y) + 2[E(X) - E(X)] \cdot [E(Y) - E(Y)]$$

$$= Var(X) + Var(Y)$$

Note: This property can be extended to any finite sum of multiple independent random variables.

Properties of the variance

5. (Chebyshev's inequality) Assume that the random variable X has mathematical expectation $E(X) = \mu$, variance $Var(X) = \sigma^2$. For any positive number ε.

Inequality $\boldsymbol{P\{|X-\mu| \geq \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}}$ holds.

Note:

- Another form of Chebysev's inequality:

$$\boldsymbol{P\{|X-\mu| < \varepsilon\} > 1 - \frac{\sigma^2}{\varepsilon^2}}$$

Properties of the variance

Example 6. Assume the mathematical expectation of X is E(X), and the variance is Var(X). **Calculate:** The mathematical expectation and variance of random variable

$$Y = \frac{X - E(X)}{\sqrt{Var(X)}}$$

.

Answer:

$$E(Y) = E\left(\frac{X - E(X)}{\sqrt{Var(X)}}\right) = \frac{1}{\sqrt{Var(X)}} E[X - E(X)]$$

$$= \frac{1}{\sqrt{Var(X)}} [E(X) - E(X)] = 0$$

$$Var(Y) = Var\left(\frac{X - E(X)}{\sqrt{Var(X)}}\right) = \frac{1}{Var(X)} Var[X - E(X)]$$

$$= \frac{Var(X)}{Var(X)} = 1$$

Y is a standardized random variable

Section 3 Covariance and correlation coefficient

- Covariance
- Correlation coefficient

Section 3 Covariance and correlation coefficient

Introduction of problems

The mathematical expectation and variance only describe a certain aspect of **a random variable**, while the covariance and correlation coefficient are numerical characteristics that describe the relationship between **two random variables.**

Covariance and correlation coefficient

The numerical characteristics of the relationship between the components of a multi-dimensional random variable.

Properties of the variance

4. If X, Y are independent random variables, then:

$$Var(X+Y) = Var(X) + Var(Y)$$

Because X and Y are independent of each other, X-E(X) and Y-E(Y) are also independent of each other

Prove: $Var(X+Y) = E[(X+Y) - E(X+Y)]^2$

$$= E\{[X - E(X)] + [Y - E(Y)]\}^2$$

$$= E[X - E(X)]^2 + E[Y - E(Y)]^2 + \underline{2E\{[X - E(X)]}\underline{[(Y - E(Y)]\}}$$

$$= Var(X) + Var(Y) \qquad +2E[X - E(X)] \cdot E[Y - E(Y)]$$

$$= Var(X) + Var(Y) + 2[E(X) - E(X)] \cdot [E(Y) - E(Y)]$$

$$= Var(X) + Var(Y)$$

Section 3 Covariance and correlation coefficient

◆ If random variables X and Y are independent,

then, $\boldsymbol{E\{[X-E(X)][(Y-E(Y)]\}=0}$

◆ If, $E(X-EX)(Y-EY)\neq 0$

then, X and Y are not independent, but have a certain relationship.

Covariance

1.Definition Suppose X and Y are random variables, then the covariance of the r.v. X and Y is

$$Cov(X,Y) = E\left[(X-\mu_X)(Y-\mu_Y)\right]$$

where $\mu_X = E(X)$ and $\mu_Y = E(Y)$ exist.

when X = Y, $Cov(X,Y) = Var(X)$,
That's means variance is a special case of covariance.

- The variance of a single r.v. has been looked upon as a measure of dispersion of the distribution of the r.v.
- Some motivation will be given subsequently to the effect that the **Cov(X,Y)** may be thought of as a measure of the degree to which **X** and **Y** tend to increase or decrease simultaneously when **Cov(X, Y) > 0** or to move toward opposite directions when **Cov(X, Y) < 0.**

Covariance

2. A formula for calculating covariance

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$

Prove:

$$Cov(X,Y) = E\{[X - E(X)][(Y - E(Y)]\}$$
$$= E(XY) - E(X)E(Y) - \cancel{E(Y)E(X)} + \cancel{E(X)E(Y)}$$
$$= E(XY) - E(X)E(Y)$$

Obviously,

if X and Y are independent to each other, then ***Cov(X, Y)= 0***

Covariance

3. Properties of covariance

$$Cov(X, X) = Var(X)$$

$$Cov(X, Y) = Cov(Y, X)$$

$$Cov(aX, bY) = abCov(X, Y)$$

$$[Cov(X, Y)]^2 = Var(X)Var(Y)$$

$$Cov(X_1 + X_2, Y) = Cov(X_1, Y) + Cov(X_2, Y)$$

Covariance

4. The relationship between the variance of the sum of random variables and the covariance

$$(1)\ \ Var(X \pm Y) = Var(X) + Var(Y) \pm 2Cov(X, Y)$$

For r.v. X and Y, the following propositions are equivalent to each other:

1) X and Y are uncorrelated

2) $\mathrm{Cov}(X, Y) = 0$

3) $\mathrm{E(XY) = E(X)E(Y)}$

4) $Var(\mathrm{X + Y}) = Var(\mathrm{X}) + Var(\mathrm{Y})$

5) $Var(\mathrm{X - Y}) = Var(\mathrm{X}) + Var(\mathrm{Y})$

6) $Var(\mathrm{X - Y}) = Var(\mathrm{X + Y})$

Covariance

4. The relationship between the variance of the sum of random variables and the covariance

$$(2)\quad Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) + 2\sum\sum_{i<j} Cov(X_i, X_j)$$

If $X_1, X_2, \ldots, X_n$ are independent to each other, then:

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var\ (X_i)$$

Covariance

New Question:

The value of the covariance reflects the relationship between X and Y to a certain extent, but it is also affected by measurement value and its units of X and Y.

$$Cov(X,Y) = E\{[X - E(X)][(Y - E(Y)]\}$$

$$Cov(aX, bY) = abCov(X,Y)$$

$$[Cov(X,Y)]^2 = Var(X)Var(Y)$$

Covariance

- From the definition of the $Cov(X, Y)$, it follows that if X is measured in units, call them a, and Y is measured in units, call them b, then $Cov(X; Y)$ is measured in units ab.
- Furthermore, the range of values of the covariance $Cov(X; Y)$ of two r.v.'s may vary from $-\infty$ to $+\infty$.

➢ These two characteristics of a covariance are rather undesirable and are both eliminated through the standardization process of replacing X and Y by $\frac{X-\mu_X}{\sigma_X}$ and $\frac{Y-\mu_Y}{\sigma_Y}$. This covariance is called the **correlation coefficient** of the r.v.'s X and Y.

Section 3 Covariance and correlation coefficient

- Covariance
- Correlation coefficient

Correlation coefficient

1. Definition The correlation coefficient of the r.v. X and Y is

$$\rho_{XY} = \rho(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

$$= \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}}$$

where $\mu_X = E(X), \mu_Y = E(Y), \sigma_X = \sqrt{Var(X)}$ and $\sigma_Y = \sqrt{Var(Y)}$ exist.

2. Definition The r.v. X and Y are said to be uncorrelated if $\rho_{XY} = 0$.

Correlation coefficient

3. Theorem Let ρ_{XY} be the correlation coefficient of the random variables X and Y, then

(1) $\rho_{XY}^2 \leq 1$ or $|\rho_{XY}| \leq 1$;

(2) $\rho_{XY}^2 = 1$ or $|\rho_{XY}| = 1$

iff there are two constant a, b such that $Y = a + bX, (b \neq 0)$.

$P(Y = aX + b) = 1$

X and Y are linearly related with probability 1

Correlation coefficient

Prove: (1) $|\rho_{XY}| \leq 1$

From the definition of variance and covariance, for any real b, there are:

$$0 \leq Var(Y - bX) = \quad b^2 Var(X) + Var(Y) - 2bCov(X,Y)$$

Let $b = \dfrac{Cov(X,Y)}{Var(X)}$,

then : $Var(Y - bX) = Var(Y) - \dfrac{[Cov(X,Y)]^2}{Var(X)}$

$$= Var(Y)[1 - \frac{[Cov(X,Y)]^2}{Var(X)Var(Y)}] = Var(Y)[1 - \rho_{XY}{}^2]$$

since Var(Y) is positive, thus must be: $1 - \rho_{XY}{}^2 \geq 0$

so, $|\rho_{XY}| \leq 1$

Correlation coefficient

Sufficiency

Prove: $(2)\ |\rho_{XY}| = 1 \quad \Leftrightarrow$ There are constants a, b ,and

$$P(Y = aX + b) = 1$$

$$Var(X \pm Y) = Var(X) + Var(Y) \pm 2Cov(X, Y)$$

$$Var\left(\frac{X - E(X)}{\sqrt{Var(X)}} \pm \frac{Y - E(Y)}{\sqrt{Var(Y)}}\right) = Var\left(\frac{X - E(X)}{\sqrt{Var(X)}}\right) + Var\left(\frac{Y - E(Y)}{\sqrt{Var(Y)}}\right)$$

$$\pm 2Cov\left(\frac{X - E(X)}{\sqrt{Var(X)}}, \frac{Y - E(Y)}{\sqrt{Var(Y)}}\right)$$

$$= 1 + 1 \pm 2\rho_{XY} = 2(1 \pm \rho_{XY})$$

$\frac{X - E(X)}{\sqrt{Var(X)}}$ and $\frac{Y - E(Y)}{\sqrt{Var(Y)}}$

are standardized random variables, so its mean is 0 and variance is 1.

Correlation coefficient

Sufficiency

1) When $\rho_{XY} = 1$,

$$Var\left(\frac{X - E(X)}{\sqrt{Var(X)}} - \frac{Y - E(Y)}{\sqrt{Var(Y)}}\right) = 0$$

From the probability of variance, we know:

$$P\left(\frac{X - E(X)}{\sqrt{Var(X)}} - \frac{Y - E(Y)}{\sqrt{Var(Y)}} = c\right) = 1$$ where c is a constant

Thus, $P\ (Y = aX + b) = 1$

with $a = \frac{\sqrt{Var(Y)}}{\sqrt{Var(X)}}$, $b = E(Y) - \sqrt{Var(Y)}\left(\frac{E(X)}{\sqrt{Var(X)}} + c\right)$

2) When $\rho_{XY} = -1$, this conclusion can be derived in the same way.

Correlation coefficient

Necessity

Prove: $(2)\ |\rho_{XY}| = 1$

There are constants a, b ,and $P(Y = aX + b) = 1$ $\Leftrightarrow$ $|\rho_{XY}| = 1$

$$P(Y = aX + b) = 1, \quad so \quad P(Y - aX = b) = 1$$

$$\underline{Var(Y - aX) = 0}$$

$$Var(Y - aX) = Var(Y) + a^2 Var(X) - 2Cov(Y, aX)$$

$$= Var(aX + b) + a^2 Var(X) - 2E[Y - E(Y)][aX - aE(X)]$$

$$= \underline{a^2 Var(X) + a^2 Var(X)} \underline{- 2aE[X - E(X)][Y - E(Y)]}$$

$$= 2a^2 Var(X) - 2aCov(X, Y) = 0$$

Covariance definition

So, $\quad Cov(X, Y) = aVar(X)$

Correlation coefficient

$$Cov(X,Y) = aVar(X)$$

$$= a\sqrt{Var(X)}\sqrt{Var(\frac{Y-b}{a})} \quad = a\sqrt{Var(X)}\sqrt{\frac{Var(Y)}{a^2}}$$

$$= \frac{a}{|a|}\sqrt{Var(X)}\sqrt{Var(Y)}$$

$$Y = aX + b$$

So, $|Cov(X,Y)| = \left|\sqrt{Var(X)}\sqrt{Var(Y)}\right|$

$$|\rho_{XY}| = 1$$

Correlation coefficient

$$Cov(X,Y) = E\{[X - E(X)][(Y - E(Y)]\}$$
$$= E(XY) - E(X)E(Y)$$

X and Y are independent $\Rightarrow$ X and Y are uncorrelated

$\not\Leftarrow$

$E(XY) = E(X)E(Y)$ $\qquad$ $\rho_{XY} = 0$

- When X and Y are independent of each other, $\rho_{XY} = 0$, but its inverse is not true.
- $\rho_{XY} = 0$ does not means that X and Y are independent of each other.

Correlation coefficient

Example 1.

Suppose that X and Y obey a uniform distribution on $x^2 + y^2 \leq 1$,

$$f(x,y) = \begin{cases} \dfrac{1}{\pi} & x^2 + y^2 \leq 1 \\ 0 & x^2 + y^2 > 1 \end{cases}$$

Verification X and Y are not related, but not independent of each other.

Prove:

It is known that the marginal pdf of X, Y is:

$$f_X(x) = \begin{cases} \dfrac{2}{\pi}\sqrt{1-x^2}, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases} \qquad f_Y(y) = \begin{cases} \dfrac{2}{\pi}\sqrt{1-y^2}, & |y| \leq 1 \\ 0, & |y| > 1 \end{cases}$$

Correlation coefficient

Obviously, $f(x,y) \neq f_X(x) f_Y(y)$

Therefore, X and Y are not independent.

Odd function

$$E(X) = \int_{-1}^{1} x \frac{2}{\pi} \sqrt{1-x^2} \ dx = 0$$

$$E(Y) = \int_{-1}^{1} y \frac{2}{\pi} \sqrt{1-y^2} \ dy = 0$$

$$E(XY) = \iint_{x^2+y^2 \leq 1} xy \frac{1}{\pi} \ dxd \quad = \frac{1}{\pi} \int_{-1}^{1} dx \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} xydy \quad = 0$$

Thus: $Cov(X,Y) = E(XY) - E(X)E(Y) = 0$

So , $\rho_{xy} = 0$, and X, Y are not related.

Correlation coefficient

- When $\rho_{XY} = 0$, X and Y are uncorrelated.

  if X and Y are independent, then X uncorrelated to Y; but the opposite is not true.

- Only if in the following situations, independent and uncorrelated are equivalent.

  If (X, Y) obeys two-dimensional normal distribution, then:

  X and Y are independent $\Leftrightarrow$ X and Y are uncorrelated.

Correlation coefficient

Example 2. Assuming(X, Y) obey a two-dimensional normal distribution, and its probability density is:

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}[\frac{(x-\mu_1)^2}{{\sigma_1}^2}-2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{{\sigma_2}^2}]}$$

Question: the correlation coefficient between X and Y ?

Answer: From the known, the marginal probability density of X, Y is:

$$f_X(x)=\frac{1}{\sqrt{2\pi}\sigma_1}e^{-\frac{(x-\mu_1)^2}{2{\sigma_1}^2}},\quad -\infty<x<+\infty$$

$$f_Y(y)=\frac{1}{\sqrt{2\pi}\sigma_2}e^{-\frac{(y-\mu_2)^2}{2{\sigma_2}^2}},\quad -\infty<y<+\infty$$

$$E(X)=\mu_1,\quad E(Y)=\mu_2,\quad Var(X)={\sigma_1}^2,\quad Var(Y)={\sigma_2}^2$$

Correlation coefficient

Thus, $$Cov(X,Y) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} (x-\mu_1)(y-\mu_2) f(x,y)\,dxdy$$

$$= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} (x-\mu_1)(y-\mu_2) \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{{\sigma_1}^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{{\sigma_2}^2}\right]} dxdy$$

$$= \rho \cdot \sigma_1 \cdot \sigma_2$$

So, $$\rho_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(Y)}} = \rho$$

Correlation coefficient

Conclusion:

The meanings of the five parameters in the probability density function of the two-dimensional normal distribution are:

1. the mathematical expectation of X and Y;
2. the variance of X and Y;
3. and the correlation coefficient of X and Y,

The two-dimensional normal distribution is completely determined by these five parameters.

Correlation coefficient

Note:

- The correlation coefficient describes the degree of **"linear correlation"** between X and Y.
- If the linear function of X (a+bX) is considered to approximate Y; and $e = E\{[Y - (a + bX)]^2\}$ is used to measure the quality of this approximate result. Then, the smaller the value of e, the better the similarity between $a + bX$ and Y.

Correlation coefficient

Now use the extremum method in calculus to find a and b , in where e is minimized:

$$e = E\{[Y-(a+bX)]^2\}$$

$$= E(Y^2) + b^2 E(X^2) + a^2 - 2b\,E(XY) + 2ab\,E(X) - 2aE(Y)$$

Let:
$$\begin{cases} \dfrac{\partial e}{\partial a} = 2a + 2bE(X) - 2E(Y) = 0 \\ \dfrac{\partial e}{\partial b} = 2bE(X^2) - 2E(XY) + 2aE(X) = 0 \end{cases}$$

Then ,
$$\begin{cases} b_0 = \dfrac{Cov(X,Y)}{Var(X)} \\ a_0 = E(Y) - b_0 E(X) \end{cases}$$

The best approximation:

$L(X)=a_0+b_0X$

Correlation coefficient

Then, the remainder of this approximation is:

$$E\{[Y - L(X)]^2\} = Var(Y)(1 - \rho_{XY}{}^2)$$

Thus,

if $\rho_{XY} = \pm 1$, then Y and X have a strict linear relationship;

else if $\rho_{XY} = 0$, then Y has no linear relationship with X;

else if $0 \leq |\rho_{XY}| \leq 1$, then :

- The closer $|\rho_{XY}|$ **is to 1**, the higher the degree of linear correlation between Y and X;
- The closer $|\rho_{XY}|$ is to 0, the weaker the linear correlation between Y and X.